

Signals, Systems and Transforms

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Chapter 1

Overview of Digital Signal Processing

Chapter Intended Learning Outcomes:

- i) Understand basic terminology in digital signal processing
- ii) Differentiate digital signal processing and analog signal processing
- iii) Describe basic digital signal processing application areas

Signal

- Anything that conveys information, e.g.,
 - Speech
 - Electrocardiogram (ECG)
 - Radar pulse
 - DNA sequence
 - Stock price
 - Image
 - Video

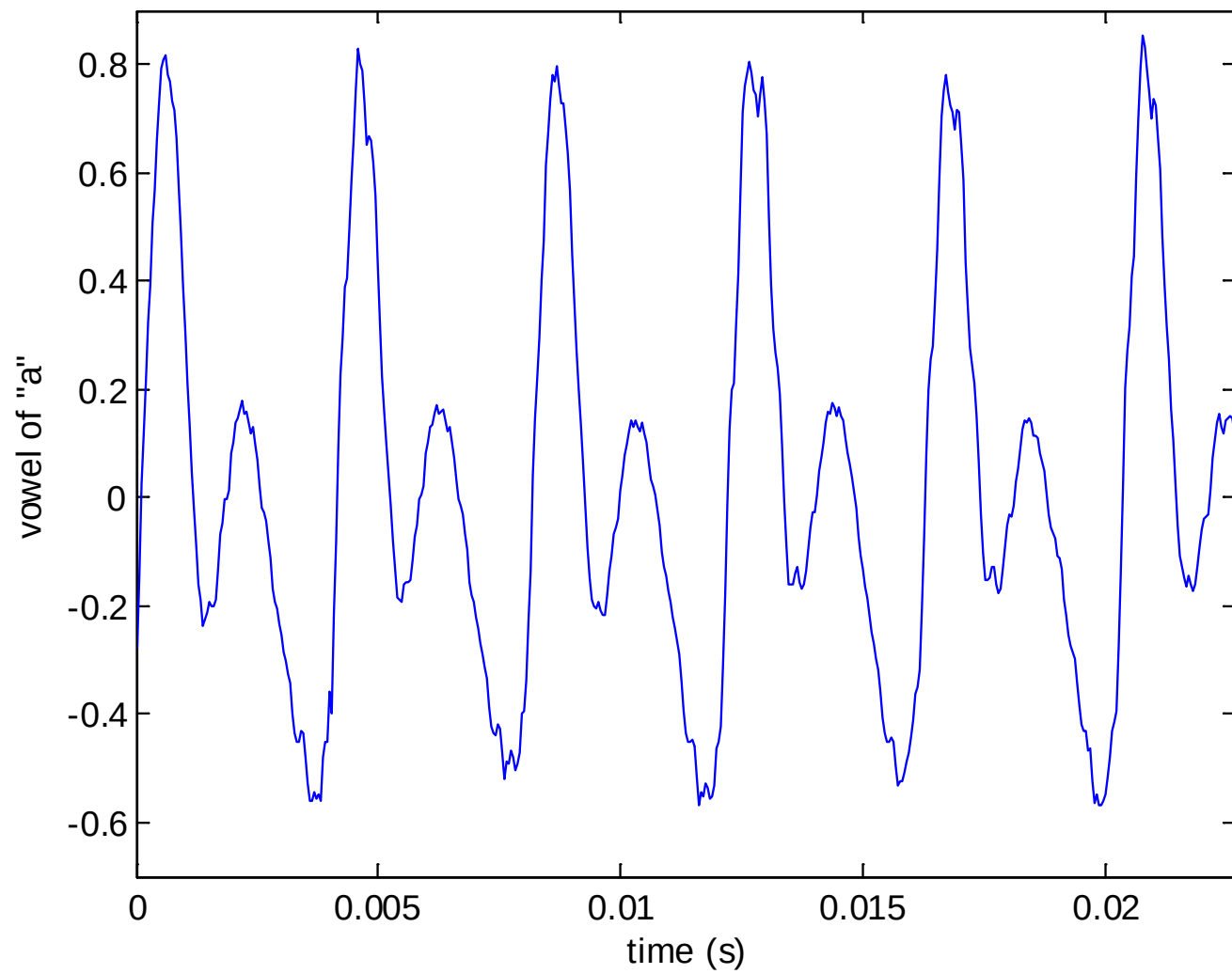


Fig. 1.1 Speech

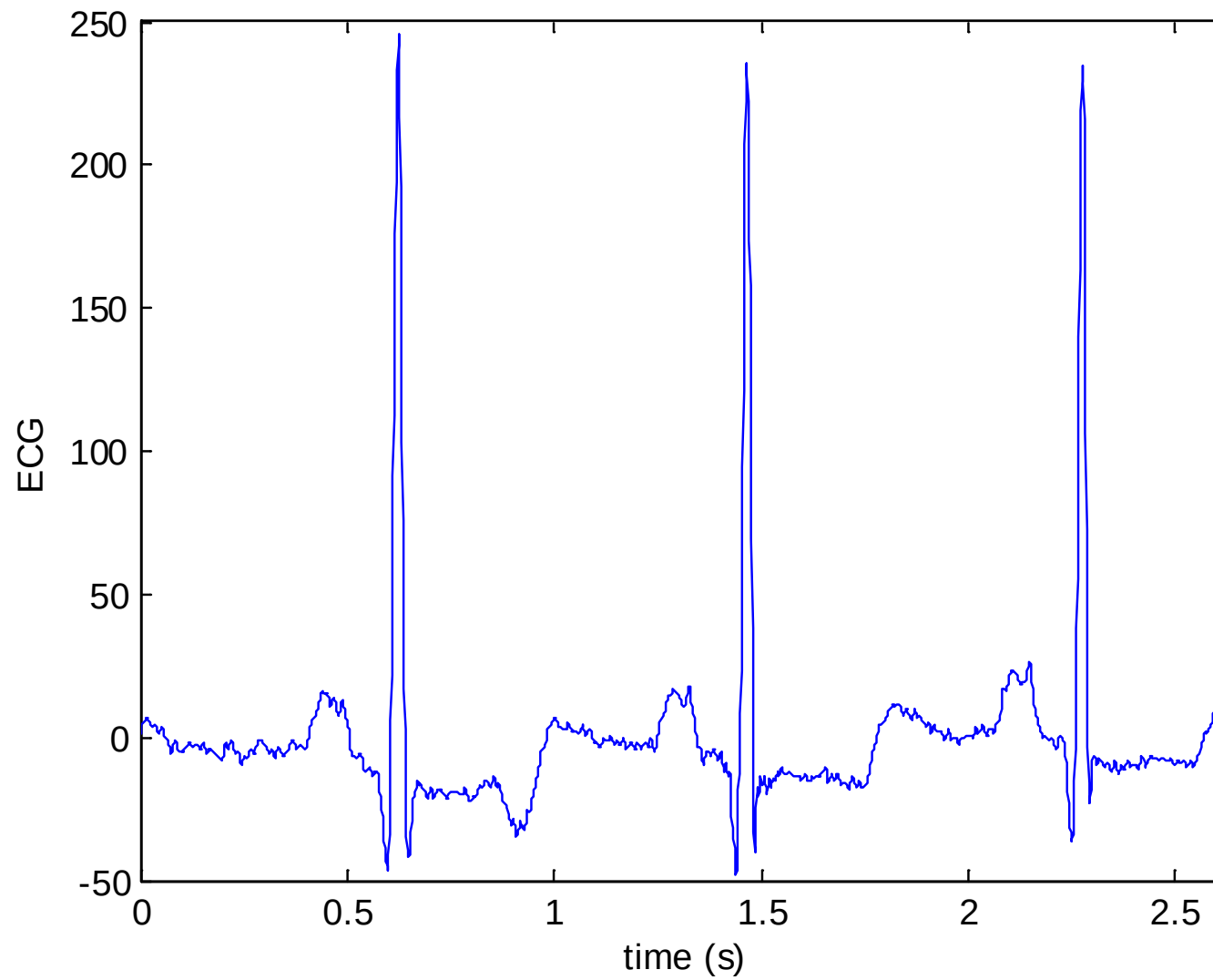


Fig. 1.2 ECG

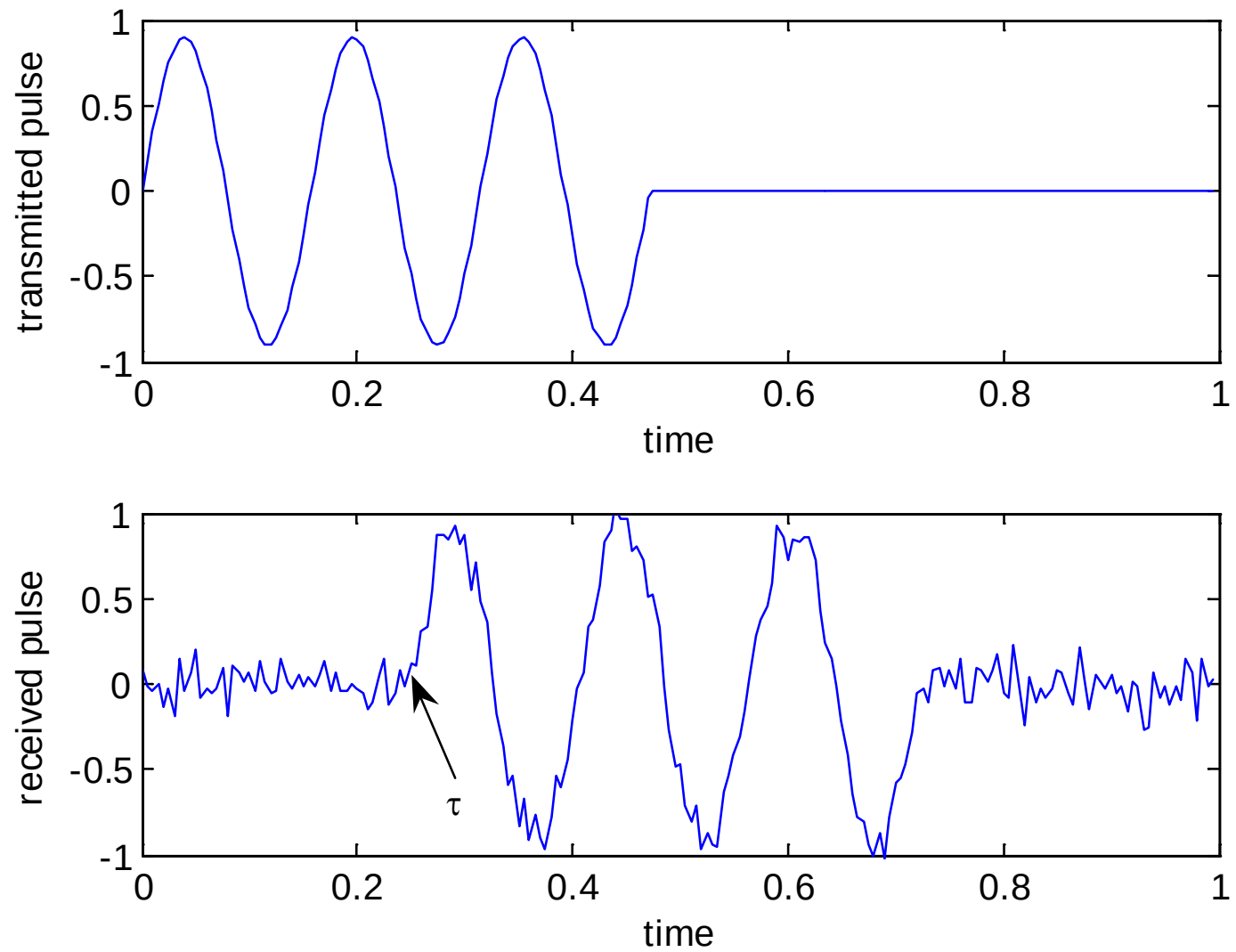


Fig. 1.3 Transmitted & received radar waveforms

- Can be a function of one, two or three independent variables, e.g., speech is 1-D signal, function of time; image is 2-D, function of space; wind is 3-D, function of latitude, longitude and elevation
- 3 types of signals that are functions of **time**:
 - **Continuous-time** (analog) $x(t)$: defined on a continuous range of time t , amplitude can be any value
 - **Discrete-time** time $x(nT)$: defined only at discrete instants of $t = \dots - T, 0, T, 2T, \dots$, amplitude can be any value
 - **Digital** (quantized) $x_Q(nT)$: both time and amplitude are discrete, i.e., it is defined only at $t = \dots - T, 0, T, 2T, \dots$ and amplitude is confined to a finite set of numbers

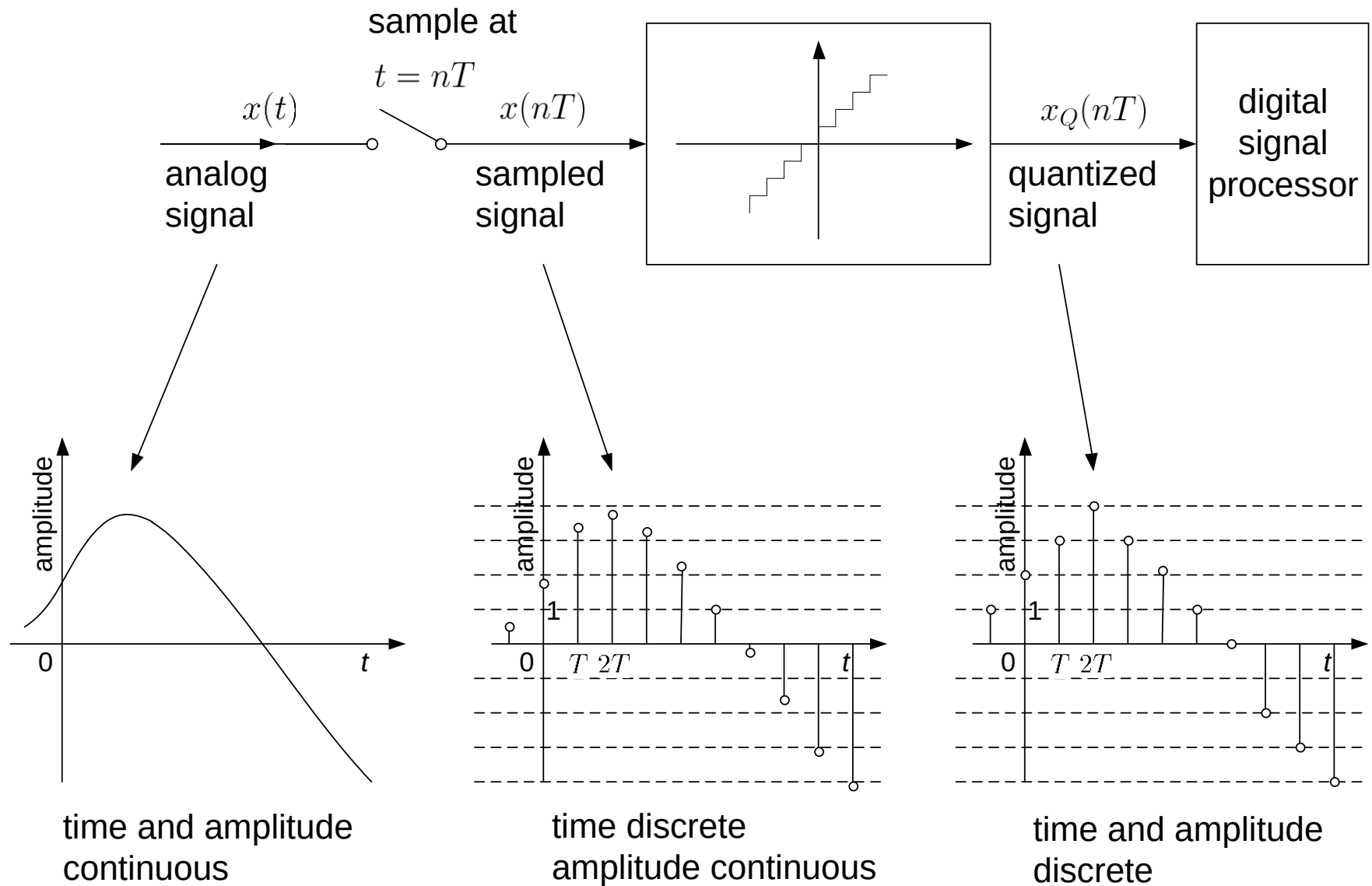


Fig. 1.4 Relationships between $x(t)$, $x(nT)$ and $x_Q(nT)$

$x(nT)$ at $n = 0$ is close to 2 and $x_Q(0) = 2$

$x(nT) \in (3, 4)$ at $n = 1$ and $x_Q(T) = 3$

Using 4-bit representation, $x_Q(0) = 0010$ and $x_Q(T) = 0011$, and in general, the value of $x_Q(nT)$ is restricted to be an integer between -8 and 7 according to the two's complement representation.

In digital signal processing (DSP), we deal with $x_Q(nT)$ as it corresponds to computer-based processing. Throughout the course, it is assumed that **discrete-time signal = digital signal**, or the quantizer has infinite resolution.

System

- Mathematical model or abstraction of a physical process that relates input to output, e.g.,
 - Grading system: inputs are coursework and examination marks, output is grade
 - Squaring system: input is 5, then the output is 25
 - Amplifier: input is $\cos(\omega t)$, then output is $10\cos(\omega t)$
 - Communication system: input to mobile phone is voice, output from mobile phone is CDMA signal
 - Noise reduction system: input is a noisy speech, output is a noise-reduced speech
 - Feature extraction system: input is $\cos(\omega t)$, output is ω
- Any system that processes digital signals is called a digital system, digital filter or digital (signal) processor

Processing

- Perform a particular function by passing a signal through system

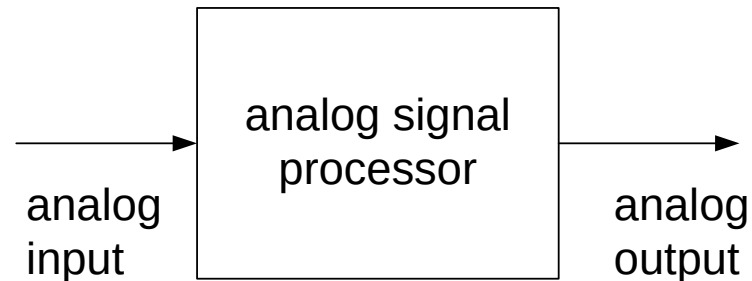


Fig. 1.5 Analog processing of analog signal



Fig. 1.6 Digital processing of analog signal

Advantages of DSP over Analog Signal Processing

- Allow development with the use of PC, e.g., MATLAB
- Allow **flexibility** in reconfiguring the DSP operations simply by changing the program
- **Reliable**: processing of 0 and 1 is almost immune to noise and data are easily stored without deterioration
- **Lower cost** due to advancement of VLSI technology
- **Security** can be introduced by encrypting/scrambling
- **Simple**: additions and multiplications are main operations

DSP Application Areas

- Speech

- Compression (e.g., Linear Predictive Coding (LPC) is a coding standard for compression of speech data)
- Synthesis (computer production of speech signals, e.g., text-to-speech engine by Microsoft)
- Recognition (e.g., automatic telephone number enquiry system)
- Enhancement (e.g., noise reduction for a noisy speech)

- Audio

- Compression (e.g., MP3 is a coding standard for compression of audio data)

- Generation of music by different musical instruments such as piano, cello, guitar and flute using computer.
- Song with low-cost electronic piano keyboard quality.
- Automatic music transcription (writing a piece of music down from a recording)

- Image and Video

- Compression (e.g., JPEG and MPEG is are coding standards for image and video compression, respectively)
- Recognition such as face, palm and fingerprint
- Enhancement
- Construction of 3-D objects from 2-D images

- **Communications:** encoding and decoding of digital communication signals
- **Astronomy:** finding the periods of orbits
- **Biomedical Engineering:** medical care and diagnosis, analysis of ECG, electroencephalogram (EEG), nuclear magnetic resonance (NMR) data
- **Bioinformatics:** DNA sequence analysis, extracting, processing, and interpreting the information contained in genomic and proteomic data
- **Finance:** market risk management, trading algorithm design, investment portfolio analysis